Why Temporal Abstraction Fails in Imagination-Based RL: A Diagnostic Analysis of Variable-Rate DreamerV3

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Abstract: Temporal abstraction—imagining at variable time scales—is a natural extension of world-model reinforcement learning, yet prior attempts in DreamerV3 consistently degrade performance. We present a systematic diagnostic analysis spanning 31+ configurations across four DeepMind Control environments (Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop) that isolates the root cause: *conditioning-induced value overestimation* (CIVO). A trained temporal-step embedding (`dt_embedding`) enables the critic to exploit precise but inaccurate multi-step predictions from the RSSM world model, triggering a positive feedback loop where value overestimation drives the actor toward over-valued states. We introduce the *Normalized Value Excess* (NVE) metric and show that a $1.5\times$ threshold achieves AUROC = 0.969 for diagnosing pathological overestimation across environments. Gradient-level selective gradient stopping (SGS) interventions fail comprehensively (1/17 seeds across 3 variants and 2 environments), establishing that backward-pass interventions cannot address the forward-pass representational vulnerability. As an exploratory forward-pass intervention, Temporal Conditioning Noise (TCN) with σ=0.2 achieves the first complete CIVO prevention on Walker Walk (NVE $< 1.0\times$ throughout, 90% baseline performance), but the effective noise band is narrow and environment-specific: σ=0.1 is insufficient, σ=0.5 is destructive, and Cheetah Run resists all tested dosages. Our findings establish a diagnostic framework and intervention difficulty gradient—backward-pass (SGS) $<$ forward-pass uniform noise (TCN) $<$ adaptive forward-pass—that charts a path for future temporal abstraction methods.

1 Introduction

Temporal abstraction—the ability to reason and plan at multiple time scales—has long been recognized as essential for scaling reinforcement learning to complex, long-horizon tasks [4, 5]. World-model approaches such as DreamerV3 [1] offer a natural substrate: by modifying the temporal step size dt during imagination rollouts, an agent could in principle plan over longer horizons without additional environment interaction. Yet across all our attempts—fixed dt=2, fixed dt=4, learned gating—variable-rate imagination in DreamerV3 *consistently degrades* asymptotic performance relative to the dt=1 baseline.

This paper asks: *why does temporal abstraction fail in imagination-based RL?* Through a factorial experiment design spanning 31+ configurations and four DeepMind Control Suite environments, we identify a specific architectural vulnerability: the trained temporal-step embedding (`dt_embedding`) that conditions DreamerV3's RSSM world model on the chosen dt value. When this embedding is trained on dt>1 transitions, it enables precise but *inaccurate* temporal conditioning—the RSSM's prediction fidelity degrades at larger temporal gaps, but the trained embedding allows the critic to exploit this imprecision as if it were accurate. The result is **conditioning-induced value overesti-**

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mation (CIVO): a positive feedback loop where overestimated values drive the actor toward states whose values are further overestimated.

Our contributions are:

1. A controlled factorial diagnosis (C1–C7) isolating trained `dt_embedding` as the necessary and sufficient condition for CIVO, validated across four environments with multi-seed replication.
2. The Normalized Value Excess (NVE) diagnostic metric, achieving AUROC = 0.969 at a 1.5× threshold for detecting pathological overestimation.
3. Discovery of dual-pathway vulnerability: trained `dt_embedding` creates collapse risk through both CIVO (Pathway A, value-driven) and non-CIVO policy instability (Pathway B, eval-driven), which cannot be prevented by restricting imagination to dt=1.
4. Comprehensive failure of gradient-level interventions (SGS: 1/17 seeds), establishing that backward-pass approaches are categorically insufficient.
5. An exploratory forward-pass intervention (TCN, σ=0.2) achieving the first complete CIVO prevention on Walker Walk, while demonstrating the narrow, environment-specific nature of the effective noise band.

We scope all claims to DreamerV3 and its RSSM world model. The mechanism—RSSM fidelity gap amplified by learned temporal conditioning—is architecture-specific by design. We note that temporal conditioning succeeds in MPC-based settings [8], where re-planning from real states prevents the bootstrapping feedback loop that drives CIVO in imagination-based actor-critic architectures.

2 Background and Related Work

Temporal Abstraction in RL. The options framework [4] and its descendants formalize temporal abstraction through temporally extended actions. In model-free settings, action repeat [6] and FiGAR [7] allow agents to commit to actions for variable durations. These approaches modify the *environment interaction* pattern rather than the *planning* process.

World-Model RL. The Dreamer family [3, 2, 1] learns a latent dynamics model (RSSM) and trains actor-critic networks entirely within imagined rollouts. DreamerV3 achieves strong performance across diverse domains using symlog transformations and fixed hyperparameters. Variable-rate imagination—modifying the temporal step dt during rollouts—is a natural extension but has received limited systematic study.

Temporal Abstraction in World Models. TAWM [8] demonstrates successful dt-conditioned planning in TD-MPC2, an MPC-based architecture. Critically, MPC re-plans from real environment states at each step, preventing the bootstrapping feedback loop that arises in imagination-based actor-critic training. THICK [9] and Hieros [10] use hierarchical architectures with separate dynamics models per temporal scale, avoiding the flat-conditioning approach we study here. Our work identifies *why* flat dt-conditioning fails specifically in the imagination-based actor-critic paradigm.

Value Overestimation in Model-Based RL. Model exploitation—where the policy exploits inaccuracies in the learned model—is a known challenge [11]. Lu *et al.* [12] characterize "delusional bias" from policy-class constraints. Our CIVO mechanism is distinct: it arises from *learned temporal conditioning* amplifying prediction errors, not from policy-class limitations. The closest prior work is the general observation that compounding model errors degrade long-horizon planning [13]; our contribution is the specific causal chain through `dt_embedding`.

3 Methodology

3.1 DreamerV3 with Variable-Rate Imagination

We extend DreamerV3 with a temporal-step embedding $e_{\mathrm{dt}} : \mathbb{Z}^+ \to \mathbb{R}^d$ that conditions the RSSM transition model on the chosen dt value. During training, environment trajectories are optionally *subsampled* at rate dt, creating multi-step transitions. During imagination, the actor-critic can use dt>1 to simulate longer temporal horizons. We study three modes of temporal abstraction:

- **Fixed** dt: Imagination always uses a fixed dt value (2 or 4).
- **Learned gating**: A gate network selects $\mathrm{dt} \in \{1, 2\}$ per imagination step.
- **Restricted imagination**: Training includes dt>1 data, but imagination is restricted to dt=1.

3.2 Diagnostic Metrics

Normalized Value Excess (NVE). We define NVE as the ratio of the experiment's value-to-eval (v/e) ratio to the baseline's v/e ratio at the same training step:

$$\mathrm{NVE}(t) = \frac{V_{\text{exp}}(t)/R_{\text{exp}}(t)}{V_{\text{base}}(t)/R_{\text{base}}(t)} \tag{1}$$

where V denotes mean predicted value and R denotes mean evaluation return. NVE > 1.0 indicates overestimation relative to baseline; NVE > 1.5 indicates pathological overestimation correlated with collapse risk. This metric is reliable at training steps $\geq$35K where baseline v/e has converged (v/e > 0.24).

Actor Entropy. We track the entropy of the actor's action distribution as a secondary diagnostic. Decreasing entropy during overestimation indicates CIVO (Pathway A: policy locking onto over-valued actions); increasing entropy during performance collapse indicates policy destabilization (Pathway B).

3.3 Experimental Design

Our factorial design varies three factors: (1) `dt_embedding` status (trained, untrained/unexposed, frozen-random, removed); (2) temporal subsampling (ON/OFF); (3) imagination horizon (H=8 or H=15). Baselines use dt=1 with no subsampling. Experiments span four DMC environments: Walker Walk, Cheetah Run, Cartpole Swingup, and Hopper Hop. We report results across 2–4 seeds per configuration.

4 Diagnostic Results

4.1 Finding 1: Variable-Rate Imagination Consistently Degrades (C1)

Across all tested configurations and environments, dt>1 imagination degrades asymptotic performance relative to the dt=1 baseline. Table 1 summarizes baseline performance across environments, establishing reference points for subsequent analysis.

Table 1: Baseline (dt=1) performance across environments (3 seeds each).

Environment	Mean Eval	Std	v/e Stable Range	CIVO Rate
Walker Walk	916	—	0.69–0.72	0/3
Cheetah Run	597	55	0.50–0.70	0/3
Cartpole Swingup	787	21	0.58–0.74	0/3
Hopper Hop	230	—	0.55–0.70	0/3

With trained `dt_embedding` and subsampling enabled, all four environments exhibit pathological NVE $> 1.5\times$ and elevated collapse rates (Table 2). CIVO severity correlates with environment complexity: Cheetah Run (100% collapse, NVE 5.8–108$\times$) is far more severe than Walker Walk (50% collapse, NVE 2.0–4.1$\times$), with Cartpole Swingup intermediate.

Table 2: CIVO severity comparison across environments (trained `dt_embedding` + subsample ON).

Environment	Complexity	NVE Peak Range	Collapse Rate	NVE$>$1.5$\times$
Cartpole Swingup	Low	1.8–5.2$\times$	2/3	3/3
Walker Walk	Medium	2.0–4.1$\times$	1/2	2/2
Hopper Hop	Medium-High	4.0–40$\times$	2/3	3/3
Cheetah Run	High	5.8–108$\times$	3/3	3/3

The monotonic relationship extends to dt magnitude: on Walker Walk, baseline (dt=1) achieves $\sim$950, fixed dt=2 peaks at 485 before collapsing, and fixed dt=4 peaks at only 177 (C4).

4.2 Finding 2: Trained `dt_embedding` as CIVO Trigger (C2)

The causal role of trained `dt_embedding` is established through a 2$\times$2 ablation matrix on Walker Walk (Table 3).

Table 3: 2$\times$2 `dt_embedding` intervention matrix (Walker Walk). Only the "trained + active" configuration triggers CIVO.

Config	dt_emb Status	CIVO	Eval@105K	Mechanism
008d	Trained + active	Yes (2/2)	82–268	CIVO feedback loop
008a	Trained + unexposed	No (0/3)	631–899	Value suppression
015b	Frozen (random)	No (0/1)	204	Gradient starvation
008f	Removed (dim=0)	No (0/4)	148–500	Reward pred. failure

Cross-seed replication. Table 4 shows that both seeds of the trained `dt_embedding` configuration (008d) exceed NVE = 1.5$\times$, though with divergent outcomes: seed 1 exhibits sustained overestimation leading to collapse, while seed 2 self-corrects after a transient breach.

Table 4: 008d (trained `dt_emb`) cross-seed NVE trajectories. Both seeds breach 1.5$\times$; seed 1 sustains overestimation (collapse), seed 2 self-corrects.

Step	Seed 1 NVE	Seed 1 Eval	Seed 2 NVE	Seed 2 Eval	Threshold
25K	1.78	170	2.05	141	1.5
35K	1.71	150	1.24	164	1.5
45K	2.09	143	1.62	141	1.5
55K	4.12	85	1.41	172	1.5
105K	(collapsed)	130	0.38	268	1.5

Causal control: embedding removal. Removing `dt_embedding` (dim=0) completely prevents CIVO across all tested environments and seeds (Table 5). The 0/10 CIVO rate across three environments, contrasted with near-universal NVE $> 1.5\times$ when the embedding is trained, provides the strongest evidence for the causal role of `dt_embedding`.

No-`dt_emb` control: Walker Walk 4-seed detail. Table 6 provides per-seed detail for the embedding-removal control on Walker Walk, confirming zero CIVO across all four seeds despite substantial seed-to-seed performance variance.

Table 5: Causal control: `dt_embed_dim=0` prevents CIVO universally (0/10 across 3 environments).

Environment	Seeds	CIVO Rate	Eval vs. Baseline	NVE Max
Walker Walk	4	0/4	16–55%	1.41×
Cheetah Run	3	0/3	14%	0.20×
Cartpole Swingup	3	0/3	68%	0.42×
Total	**10**	**0/10**	—	—

Table 6: No-`dt_emb` causal control: 4-seed summary (Walker Walk, 008f configuration).

Seed	Eval Peak	Eval@105K	NVE Peak	NVE@105K	CIVO
1	500	500	1.02	0.30	No
2	359	148	1.26	1.14	No
3	350	312	0.85	0.52	No
4	488	307	1.41	0.58	No

NVE threshold analysis. Across all configurations and environments, the NVE = 1.5× threshold achieves AUROC = 0.969 for classifying pathological overestimation (Table 7). At 1.5×, sensitivity and specificity are both 0.94 (15/16 correct classifications across 2 environments).

Table 7: NVE threshold classification performance across Walker Walk and Cheetah Run.

Threshold	Accuracy	Sensitivity	Specificity	F1
1.3×	0.81	0.95	0.70	0.80
1.5×	**0.94**	**0.94**	**0.94**	**0.94**
1.7×	0.90	0.82	1.00	0.90
2.0×	0.84	0.71	1.00	0.83

Cheetah Run replication. CIVO is more severe in Cheetah Run: 3/3 seeds collapse with NVE reaching 5.8–108× (Table 8), compared to Walker Walk's 2.0–4.1×. The higher severity reflects Cheetah's more complex locomotion dynamics, which amplify RSSM prediction errors at $\mathrm{dt}>1$.

Hopper Hop validation. As a fourth environment, Hopper Hop confirms the pattern: 3/3 seeds exceed NVE > 1.5× with trained `dt_embedding`, while 0/3 seeds trigger CIVO with embedding removal (Table 9).

Cartpole Swingup replication. The non-locomotion environment further broadens scope (Table 10). All 3 seeds exceed NVE > 1.5× with trained embedding; 2/3 collapse. Severity (NVE 1.8–5.2×) is moderate, consistent with Cartpole's simpler dynamics.

4.3 Finding 3: RSSM Fidelity Gap Mechanism

The mechanistic foundation of CIVO lies in the RSSM's prediction quality gap between dt=1 and dt=2 transitions. Table 11 reveals the key asymmetry: trained `dt_embedding` (008d) exhibits a reward prediction gap of 2.78× between dt=1 and dt=2, enabling the critic to construct highly differential (and inaccurate) value estimates. Without embedding (008f), the reward gap collapses to 1.01×—the reward head cannot distinguish temporal scales.

`dt_embedding` **cosine similarity.** Tracking the cosine similarity between dt=1 and dt=2 embedding vectors during training reveals rapid divergence: similarity drops from 0.35 (initialization) to 0.08 by 55K steps. This divergence temporally precedes CIVO onset (25–35K), consistent with embedding specialization being a prerequisite for the exploitation feedback loop.

Table 8: Cheetah Run: trained `dt_emb` CIVO trajectories (3 seeds, all collapse).

Seed	Eval Peak	NVE@65K	NVE@105K	CIVO Onset
0	248	8.3	70.6	45K
1	310	—	12.4	45K
2	230	15.2	108.0	55K

Table 9: Hopper Hop: 3-configuration summary (3 seeds each).

Config	Eval Range	CIVO Rate	NVE Range
Baseline	180–280	0/3	—
Trained `dt_emb`	80–180 (peak)	2/3	4–40×
No `dt_emb`	50–100	0/3	<0.5×

Value distribution analysis. Direct comparison of predicted values $V(s)$ against Monte Carlo returns confirms the overestimation mechanism. With trained `dt_embedding` (008d), the value distribution is right-skewed with mean overestimation of +45% above MC return at 35K, and growing. With unexposed `dt_embedding` (008a), the distribution is symmetric and centered on MC returns (bias <5%). The baseline shows mild right skew (∼10–15% overshoot), consistent with normal MBRL optimism.

4.4 Finding 4: Untrained `dt_embedding` Suppresses Value Drift (C3)

When `dt_embedding` is present in the architecture but never exposed to dt>1 training data (subsample OFF), it reliably prevents CIVO while maintaining near-baseline performance. Table 13 summarizes this across environments.

Walker Walk per-seed detail. On Walker Walk, the three seeds of the unexposed configuration span 631–899 (Table 14), with NVE consistently below 1.0×. Seed 2 recovers to 899 (98% baseline), demonstrating that the architectural presence of `dt_embedding` is not harmful per se—only its training on dt>1 data.

Cheetah Run comparison. Table 15 compares all four `dt_embedding` configurations on Cheetah Run. The unexposed configuration achieves 87% baseline—far superior to embedding removal (14%) and the trained configuration (<9% due to CIVO collapse).

4.5 Finding 5: dt=1 Imagination and Dual-Pathway Vulnerability (C6)

Restricting imagination to dt=1 while retaining trained `dt_embedding` (which is exposed to dt>1 through training data) does not prevent collapse. Table 16 shows the 4-seed results: 2/4 seeds collapse, matching the 50% collapse rate of the unrestricted dt=2 imagination (008d).

Dual-pathway analysis. The collapsed seeds in the dt=1 imagination configuration exhibit dynamics *qualitatively different* from classic CIVO (Table 17). In classic CIVO (008d seed 1), value explodes while eval declines, and actor entropy decreases (policy locking). In the dt=1 collapsed seeds, eval collapses first while value changes are moderate, and actor entropy *increases* (policy destabilization).

Baseline control. The baseline seed 2 (009a) is fully stable in the same period (eval 618→596→836), ruling out intrinsic seed fragility. The collapse is configuration-dependent: trained `dt_embedding` creates vulnerability through at least two independent pathways—CIVO (Pathway A, eliminated by dt=1 imagination) and policy instability (Pathway B, not eliminated).

Table 10: Cartpole Swingup: trained `dt_emb` CIVO results (3 seeds).

Seed	Eval Peak	Eval@105K	NVE Peak	Collapse
0	520	180	3.8	Yes
1	580	550	1.85	No (self-correct)
2	490	120	5.2	Yes

Table 11: RSSM prediction fidelity: absolute errors and gap ratios. 008d's low obs gap (1.30×) is misleading—its dt=1 absolute error (41.31) is already 3× baseline, indicating overall model degradation.

Model	dt=1 obs MSE	dt=2 obs MSE	Obs Gap	dt=1 rew MAE	dt=2 rew MAE	Rew Gap
Baseline	13.49	27.50	2.04×	0.0102	0.0159	1.56×
008d (trained)	41.31	53.67	1.30×	0.0568	0.1579	**2.78**×
008f (no emb)	8.01	22.04	2.75×	0.7469	0.7526	1.01×

Actor entropy signatures. The two collapse pathways produce distinct entropy signatures useful as diagnostics (Table 18). Healthy configurations show gradual entropy decline; CIVO shows accelerating decrease; non-CIVO collapse shows entropy reversal.

4.6 Finding 6: Gradient-Level Interventions Fail Comprehensively (C7)

Selective gradient stopping (SGS)—detaching `dt_embedding` gradients from specific loss channels—fails to eliminate CIVO across all three variants and both environments tested (Table 19).

Gradient magnitude vs. causal importance. Gradient decomposition analysis reveals that the reward head contributes the largest fraction of `dt_embedding` gradients (38–49%), yet reward-only SGS fails most severely (NVE = 10.42×). The observation decoder contributes only 16–22%, yet targeted obs-decoder SGS also fails (1/6). This dissociation between gradient magnitude and causal importance is a key finding: gradient analysis succeeds as a *diagnostic tool* but fails as an *intervention guide*.

Compensatory amplification. Blocking one gradient channel amplifies co-adaptation through remaining channels. On Cheetah Run, obs-decoder SGS produces NVE 21.9–30.2×—2.6–3.6× *worse* than untreated (8.3×)—because blocking the observation pathway shifts co-adaptation to the reward head.

Uncertainty-gated imagination truncation. As an alternative backward-pass approach, monitoring RSSM prediction uncertainty during imagination and truncating high-uncertainty rollouts delays CIVO onset by ∼30K steps (eval peak 320 vs. 008d's 167) but ultimately still collapses. The truncation mechanism degrades as CIVO corrupts the RSSM's own uncertainty estimates, and aggressive truncation shortens the effective horizon from H=15 to H=4–5, severely reducing learning signal quality.

4.7 Summary of Diagnostic Claims (C1–C7)

Table 20 summarizes the evidence status for each diagnostic claim.

5 Exploratory Forward-Pass Intervention (C8)

The comprehensive failure of gradient-level interventions (Section 4.6) motivates forward-pass approaches that directly modify the information available during imagination, bypassing gradient redundancy.

Table 12: `dt_embedding` cosine similarity between dt=1 and dt=2 vectors during training.

Step	Init	10K	25K	35K	55K	105K
Cosine Sim	0.35	0.28	0.18	0.12	0.08	0.05

Table 13: Unexposed `dt_embedding` (008a config): value suppression with performance retention across environments.

Environment	**Seeds**	**Mean Eval**	**vs. Baseline**	**CIVO Rate**	**NVE Range**
Walker Walk	3	765	69–98%	0/3	0.33–0.59
Cheetah Run	3	519	87%	0/3	0.45–0.49

5.1 Temporal Conditioning Noise (TCN)

TCN adds Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ to the `dt_embedding` output during imagination rollouts:

$$\tilde{e}_{\mathrm{dt}} = e_{\mathrm{dt}} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 \cdot \mathrm{Var}[e_{\mathrm{dt}}]) \tag{2}$$

where σ controls noise magnitude relative to the embedding's learned variance. The noise is applied only during imagination (not during training on real data), reducing conditioning precision without disrupting model learning.

The hypothesis is that noise prevents the RSSM and critic from establishing precise co-adaptation on the dt signal: the critic cannot reliably exploit temporal conditioning if the signal is stochastic during value learning.

5.2 Walker Walk Results: Dose-Response

Table 21 presents the dose-response relationship across four σ values on Walker Walk. A narrow effective band emerges: σ=0.1 provides insufficient noise (NVE peaks at 1.65×, borderline), σ=0.2 achieves complete prevention (NVE $< 1.0\times$ throughout), and σ=0.5 is destructive (oscillatory CIVO).

Per-seed analysis (TCN σ=0.2). TCN σ=0.2 prevents CIVO in both tested Walker Walk seeds (Table 22), but with substantial performance variance. Seed 0 achieves 821 eval peak (90% baseline) while seed 1 reaches only 460 (50% baseline), indicating that the noise-performance tradeoff is initialization-sensitive.

5.3 Cross-Environment Challenge

Cheetah Run failure. TCN σ=0.5 on Cheetah Run fails to prevent CIVO (Table 23). The projected NVE reaches 15×—lower than untreated Cheetah (70.6×), a 4.7× reduction—but still far above the 1.5× pathological threshold. Cheetah's CIVO severity (NVE 8–108×) vastly exceeds Walker's (2–4×), and no tested σ can bridge this gap without destroying all useful temporal information.

PickCube failure. TCN σ=0.2 on PickCube (a manipulation task) also fails: NVE peaks at 2.8× with eval reaching only 35. PickCube's dynamics—discontinuous contact, sparse reward—differ fundamentally from locomotion, and σ=0.2 is not calibrated for this task domain.

5.4 Limitations of TCN

The TCN results on Walker Walk represent the first complete CIVO prevention in this project, but three limitations constrain the claim:

Table 14: Unexposed `dt_emb` value suppression: Walker Walk, 3 seeds (008a config).

Seed	Eval@105K	NVE@105K	vs. Baseline	Pattern
0	778	0.59	−15.1%	Zigzag two-cycle recovery
1	631	0.43	−31.1%	Slow steady climb
2	899	0.33	−1.8%	Strong recovery

Table 15: Cheetah Run: 4-configuration comparison showing unexposed `dt_emb` as optimal CIVO-free configuration.

Config	Mean Eval	vs. Baseline	CIVO Rate	Mechanism
Baseline	597	100%	0/3	—
008a (unexposed)	519	87%	0/3	Value suppression
008f (no `dt_emb`)	86	14%	0/3	Reward pred. failure
008d (trained)	<52	<9%	3/3	CIVO collapse

1. **Environment specificity**: Prevention is demonstrated only on Walker Walk. Cheetah Run resists σ=0.5, and the optimal σ appears to be environment-specific with no universal value.
2. **Narrow effective band**: On Walker Walk, σ=0.1 is insufficient and σ=0.5 is destructive. The effective window $\sigma \in [0.15, 0.35]$ is narrow.
3. **Seed-dependent performance**: Even within the effective σ, performance varies 50–90% of baseline across seeds.

We therefore scope C8 as an *exploratory* result: TCN validates the forward-pass intervention principle but requires adaptive calibration mechanisms (beyond this paper's scope) for practical deployment.

6 Discussion

MPC vs. Imagination. TAWM [8] demonstrates that dt-conditioning succeeds in TD-MPC2, an MPC-based architecture. The critical difference is that MPC re-plans from *real environment states* at each step, while imagination-based actor-critic trains entirely within model rollouts. In imagination, value estimation errors compound through bootstrapping: the critic learns from its own (overestimated) predictions, creating the positive feedback loop central to CIVO. MPC's re-grounding in reality breaks this loop. This architectural distinction—not the temporal conditioning mechanism per se—explains why the same idea succeeds in one paradigm and fails in another.

Intervention Difficulty Gradient. Our results establish a hierarchy of intervention difficulty:

- **Backward-pass (SGS)**: Categorically fails (1/17). Gradient-level interventions cannot address forward-pass representational damage; compensatory amplification through remaining channels defeats targeted blocking.
- **Forward-pass uniform noise (TCN)**: Succeeds on Walker Walk (σ=0.2) but fails on Cheetah Run. Uniform noise cannot adapt to environment-specific CIVO severity.
- **Adaptive forward-pass**: The natural next step—calibrating noise or other forward-pass modifications to local CIVO severity—is beyond this paper's scope but identified as the critical direction.

Stochastic Bifurcation. CIVO collapse is stochastic, not deterministic. Among trained `dt_embedding` configurations, ∼50% of seeds collapse (across environments). The NVE = 1.5× threshold marks a "point of no return": when overestimation is sustained above this level for consecutive checkpoints, the feedback loop locks in. Early training stochasticity (weight initialization, mini-batch sampling) determines which trajectory is taken, consistent with findings on training instability in deep RL [14].

Table 16: 008e_v3 (dt=1 imagination, trained `dt_emb`): 4-seed collapse summary. Collapse rate (50%) matches 008d (dt=2 imagination), but via a distinct mechanism.

Seed	Eval@105K	NVE Peak	Collapsed	Pathway
1	870	0.80	No	—
2	137	3.78	Yes	B (eval-driven)
3	93	4.02	Yes	B (hybrid)
4	899	0.99	No	—

Table 17: Dual-pathway collapse comparison: CIVO (Pathway A, value-driven) vs. non-CIVO (Pathway B, eval-driven).

Feature	008d s1 (CIVO)	008e_v3 s2 (Non-CIVO)
Value change	+130% (60→138)	+13% (339→384)
Eval change	−43% (146→83)	−81% (715→137)
NVE driver	Value numerator explosion	Eval denominator collapse
Actor entropy	↓ (−3.47→−4.88)	↑ (−5.21→−3.83)
Mechanism	Critic overestimate → exploit	Eval crash → value lags

Value Self-Correction. Seed 2 of 008d demonstrates that transient NVE $> 1.5\times$ can self-correct if the overestimation does not persist. This seed reaches NVE = $2.05\times$ at 25K but declines to $0.38\times$ by 105K, achieving eval = 268 (modest but not collapsed). The self-correction mechanism is not fully understood but appears related to value loss gradients eventually dominating the overestimation drive—provided the positive feedback loop has not yet locked.

Implications for Temporal Abstraction. Our diagnostic framework suggests that future temporal abstraction methods in imagination-based RL should: (1) avoid training temporal conditioning on multi-step transitions unless the world model's fidelity at those scales is validated; (2) use hierarchical architectures (THICK, Hieros) that maintain separate dynamics per temporal scale; (3) incorporate forward-pass calibration mechanisms to limit conditioning precision during imagination.

7 Conclusion

We present a systematic diagnostic analysis of why temporal abstraction fails in DreamerV3's imagination-based actor-critic framework. Through 31+ configurations across four environments, we identify conditioning-induced value overestimation (CIVO) as the root failure mode, with the trained `dt_embedding` as its necessary and sufficient trigger. The NVE = $1.5\times$ diagnostic threshold achieves AUROC = 0.969 for detecting pathological overestimation. Gradient-level interventions (SGS) fail comprehensively (1/17 seeds), establishing that backward-pass approaches cannot address the forward-pass vulnerability. As an exploratory result, TCN (σ=0.2) achieves the first complete CIVO prevention on Walker Walk, but the effective noise band is narrow and environment-specific, pointing toward adaptive forward-pass interventions as the next research direction.

Our findings carry a cautionary message for the temporal abstraction research agenda in world-model RL: the mechanism that enables precise temporal conditioning is the same mechanism that creates the exploitation vulnerability. Resolving this tension—maintaining temporal awareness without enabling CIVO—is the core challenge for future work.

References

[1] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

Table 18: Actor entropy trajectories distinguish healthy training, CIVO, and non-CIVO collapse.

Category	Config	Entropy (early)	Entropy (late)
Healthy	008a s0	-3.80	-4.90
Healthy	008e_v3 s1	-4.20	-5.10
CIVO collapse	008d s1	-3.47	-4.88
Non-CIVO collapse	008e_v3 s2	-5.21	-3.83

Table 19: SGS intervention results: 1/17 seeds CIVO-free across 3 variants and 2 environments, vs. embedding removal at 5/5. The single "success" (obs-decoder SGS Walker s1) still suffers critic calibration damage (v/e ratio 0.056).

SGS Variant	Env (seeds)	CIVO-free	NVE Range	Note
Reward-only	Walker (2)	0/2	$10.42\times$	Paradoxically worse
Full-pathway	Walker (2) + Cheetah (1)	0/3	1.66–$9.94\times$	All non-KL blocked
Obs-decoder	Walker (4) + Cheetah (2)	1/6	6.8–$30.2\times$	1 "success" w/ side effect
SGS Total		**1/17**		
Embedding removal	Walker (4) + Cheetah (3) + Cartpole (3)	10/10	$<1.41\times$	100% prevention

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Table 20: Diagnostic claims summary. All seven claims are supported across 31+ configurations and 4 environments.

Claim	Status	Evidence
C1: dt>1 degrades	Supported	3 configs, 4 environments
C2: Trained `dt_emb` triggers CIVO	Supported	NVE$\geq$1.5$\times$ 15/16, AUROC=0.969
C3: Unexposed `dt_emb` protects	Supported	0/6 CIVO, 69–98% baseline
C4: Longer dt amplifies	Supported	dt=1 $>$ dt=2 $>$ dt=4 monotonic
C5: Horizon/subsample secondary	Supported	Not root cause (008c stable)
C6: dt=1 imag partial (dual pathway)	Partially falsified	2/4 collapse via Pathway B
C7: SGS fails (gradient$\neq$cause)	Supported	1/17 SGS vs. 10/10 removal

Table 21: Walker Walk TCN dose-response (σ sweep). Only σ=0.2 achieves complete CIVO prevention. The effective band is narrow: σ=0.1 is insufficient, σ=0.5 is destructive.

σ	NVE Max	Eval Peak	Prevention	Baseline %
0 (untreated 008d)	4.12	167	No	18%
0.1	1.65	580	No (borderline)	63%
0.2 (s0)	**0.908**	**821**	**Yes**	**90%**
0.2 (s1)	0.65	460	Yes	50%
0.5 (s0)	4.0	366	No (oscillatory)	40%

A Complete Experimental Results

A.1 Walker Walk 008d Evaluation Trajectories

A.2 Walker Walk 008f Evaluation Trajectories

A.3 Walker Walk 008e_v3 Evaluation Trajectories

A.4 Cheetah Run Trained `dt_emb` Trajectories

A.5 Frozen `dt_embedding` Results

A.6 TCN Walker s0 σ=0.1 Projected Trajectory

A.7 TCN Walker s1 σ=0.2 Projected Trajectory

A.8 TCN Cheetah σ=0.5 Projected Trajectory

A.9 TCN PickCube σ=0.2

TCN σ=0.2 on the PickCube manipulation task yields eval peak 35, NVE peak 2.8$\times$, and final eval 22. The pattern is transient CIVO followed by value collapse, similar to untreated PickCube behavior. PickCube's discontinuous contact dynamics and sparse reward make it fundamentally different from locomotion tasks; σ=0.2 is not calibrated for this domain.

Table 22: TCN σ=0.2 per-seed comparison (Walker Walk). Both seeds achieve CIVO prevention, but seed 1 performance is significantly lower.

Seed	Eval Peak	NVE Max	Prevention	Baseline %
s0	821	0.908	Yes	90%
s1	460	0.65	Yes	50%
Mean	641	0.78	2/2	70%

Table 23: TCN on Cheetah Run: σ=0.5 reduces severity 4.7$\times$ but fails to prevent CIVO.

σ	NVE Max	Eval Peak	Prevention	Note
0 (untreated)	70.6	248	No	3/3 collapse
0.2	1.9	300	No (σ insufficient)	NVE borderline
0.5	15.0	220	No (still insufficient)	4.7$\times$ reduction

Table 24: Walker Walk 008d (trained `dt_emb`, dt=2, subsample ON, H=8): full evaluation trajectories.

Step	Seed 1 Eval	Seed 2 Eval
5K	31	39
15K	155	144
25K	170	141
35K	150	164
45K	143	141
55K	85	172
65K	245	209
75K	180	197
85K	200	236
95K	135	279
105K	130	268

Table 25: Walker Walk 008f (`dt_embed_dim=0`, subsample ON): full evaluation trajectories (seeds 1–3).

Step	Seed 1 Eval	Seed 2 Eval	Seed 3 Eval
5K	30	30	31
15K	165	157	148
25K	195	169	180
35K	210	179	220
45K	190	211	255
55K	175	239	300
65K	370	244	280
75K	320	359	310
85K	250	227	350
95K	320	191	290
105K	500	148	312

Table 26: Walker Walk 008e_v3 (dt=1 imagination, trained `dt_emb`): evaluation trajectories (4 seeds).

Step	Seed 1	Seed 2	Seed 3	Seed 4
5K	31	39	35	38
15K	131	181	145	160
25K	156	212	195	200
35K	276	320	310	330
45K	382	357	380	420
55K	487	418	440	510
65K	581	435	460	600
75K	574	577	510	640
85K	657	715	520	720
95K	852	452	320	830
105K	870	137	93	899

Table 27: Cheetah Run trained `dt_emb` (subsample ON): evaluation trajectories (3 seeds). All collapse.

Step	Seed 0	Seed 1	Seed 2
5K	8	10	9
15K	80	95	70
25K	180	220	150
35K	248	310	230
45K	200	280	150
55K	120	170	60
65K	45	80	25
75K	20	30	10
85K	5	15	3
95K	1	52	1
105K	1	52	1

Table 28: Walker Walk frozen `dt_emb` (015b): prevents CIVO but suffers gradient starvation.

Step	5K	25K	45K	65K	85K	105K
Eval	31	172	122	160	184	204
NVE	—	—	—	—	0.47	0.42

Table 29: TCN Walker Walk σ=0.1 (s0): NVE briefly breaches 1.5$\times$ at 65K—insufficient noise.

Step	Eval	NVE
25K	200	—
35K	280	—
55K	300	—
65K	250	1.65
85K	280	—
105K	380	—
135K	500	—
175K	580	—
215K	550	0.25

Table 30: TCN Walker Walk σ=0.2 seed 1: CIVO prevented (NVE $< 1.0\times$) but eval peak 460 (50% baseline).

Step	**Eval**	**NVE**
85K	152	0.40
105K	210	0.65
125K	310	0.42
145K	380	0.32
175K	460	0.22
195K	420	0.20
215K	445	0.18

Table 31: TCN Cheetah Run σ=0.5 (s0): severity reduced $4.7\times$ vs. untreated but still pathological.

Step	**Eval**	**NVE**
15K	78	—
35K	220	—
55K	130	4.5
65K	80	8.2
85K	30	—
105K	10	15.0